AP012292.1:2438862-2448599
Selenomonas ruminantium subsp. lactilytica TAM6421

Ribosomal_L27 Peptidase_Prp Ribosomal_L21p HTH_31 HTH_28 HTH_28 HTH_28 rve_2 rve HTH_21 MqsA_antitoxin SocA_Panacea MqsR_toxin NAD_binding_8 GLF Fic ADH_N ADH_zinc_N ? IspD Ribul_P_3_epim DUF4160

GG774808.1:95670-116270
Bacteroides sp. D22

DrsE_2 TusA Rhodanese Pyr_redox_dim Pyr_redox_2 Receiver NtrC-AAA AAA_lid_14 HTH_8 AA_kinase DUF2400 DUF2400 LRRs LRRs LRRs Big ? TPRs MqsR_toxin HTH_3 YxiJ SocA_Panacea ? Arm-DNA-bind_5 Phage_int_SAM_5 Phage_integrase HTH_18 DUF3575 DUF3868 P_gingi_FimA Fimbrillin_C

BLSA01000222.1:1-1554
Candidatus Hakubella thermalkaliphila

Glyco_hydro_130 ? ? MqsA_antitoxin

BTQY01000027.1:16641-27330
Gammaproteobacteria bacterium

Iron_traffic ATPgrasp_N ATP-grasp_4 LAL_C2 Fe-S_biosyn EptA_B_N Sulfatase CitMHS CitMHS DUF7351 MqsA_antitoxin SocA_Panacea MqsR_toxin DUF4143 AAA_14

BTTB01000015.1:335-6949
bacterium

HTH_3 MqsR_toxin MqsA_antitoxin SocA_Panacea DEAD Helicase_C DbpA RIO1 DUF1456 DUF1456 S4_2 DUF3087 ?

BTTF01000087.1:950-8323
bacterium

IGPD PNPase AICARFT_IMPCHas Cys_Met_Meta_PP PNPOx_N MqsR_toxin MqsA_antitoxin SocA_Panacea PNPOx_N DUF5615 DUF433 ?

DMDF01000075.1:36581-52816
Anaerolineaceae bacterium

HSP20 BPL_LplA_LipB BPL_C DNA_gyraseA_C DNA_gyraseA_C DNA_gyraseA_C DNA_gyraseA_C DNA_gyraseA_C DNA_gyraseA_C DNA_topoisoIV PBP_like_2 BPD_transp_1 BPD_transp_1 ABC_tran MqsR_toxin MqsA_antitoxin SocA_Panacea DUF4143 AAA_14 Amidohydro_1 SIS_2 NanE NanE EIID-AGA PTSIIB_sorb

DMDI01000194.1:94-5693
Desulfotomaculum sp.

Transposase_mut ? Cu_amine_oxidN1 Cu_amine_oxidN1 NAGPA ? MqsR_toxin HTH_31 SocA_Panacea Cu_amine_oxidN1 Cu_amine_oxidN1 ?

DMOT01000244.1:1-3974
Bacillota bacterium

ABC_tran Aminotran_5 MqsA_antitoxin SocA_Panacea MqsR_toxin

DMZY01000228.1:232-7720
candidate division WOR-3 bacterium

? MqsA_antitoxin SecB Big Big MG2 bMG5 A2M_BRD A2M bMG10

DNAN01000018.1:1-5963
Alteromonas australica

HTH_31 ? ? DUF927 HTH_17 HTH_17 MqsR_toxin MqsA_antitoxin SocA_Panacea Phage_integrase Phage_int_M

DNFN01000407.1:465-3821
Bacteroidetes bacterium

? ? PhageMin_Tail DUF2188 MqsR_toxin MqsA_antitoxin

DNOQ01000429.1:1-10431
Bacteroidales bacterium

GFO_IDH_MocA GFO_IDH_MocA_C2 GFO_IDH_MocA Amidohydro_2 AP_endonuc_2 ? ? MqsA_antitoxin SocA_Panacea GFO_IDH_MocA_C2 GFO_IDH_MocA DUF6807 ThuA Peptidase_S9 PETase

DQZX01000108.1:1-2913
Thermotoga sp.

ATP-synt_VA_C ATP-synt_ab ATP-synt_D MqsA_antitoxin SocA_Panacea MqsR_toxin

DRGI01000320.1:13-10252
Phycisphaerae bacterium

MqsA_antitoxin SocA_Panacea ? SAICAR_synt ? HEATs Trypsin PDZ_6 PDZ_2 LGT LGT Asp_decarbox Pantoate_ligase

DRKS01000100.1:20009-29036
Sedimenticola sp.

ABC_N ABC_ATPase MRB1590_C HKD HKD VIT1 DUF211 RloB ABC-ATPase MqsA_antitoxin SocA_Panacea MqsR_toxin RAMA ?

DRJC01000315.1:1-10050

Ignavibacteria bacterium

? UPF0182 CBM9_1 DUF5916 DUF5916 Radical_SAM Glyco_hydro_77 MqsR_toxin MqsA_antitoxin SocA_Panacea

DATAKL010000029.1:113-14705

Ignavibacteriales bacterium

? Methylase_S Methylase_S N6_Mtase HsdM_N HD ? MqsA_antitoxin SocA_Panacea ? ? ? Fn3-like Glyco_hydro_3_C Glyco_hydro_3 Sigma70_r2 Sigma70_r4_2 DUF6077

DATAMB010000020.1:19812-32891
Oscillospiraceae bacterium

HDPD ? SPP1_portal ? ? SecB MqsA_antitoxin MqsR_toxin Resolvase HTH_17 ? HTH_17 GP3_package NrS1-1_pol-like_HBD D5_N DUF5906 Pox_D5 ? ?

DAORMS010000016.1:45963-64367
Cyclobacteriaceae bacterium

DUF3685 P-loop_TraG CagE_TrbE_VirB DUF3875 DUF4133 DUF4134 ? ? ? ? MqsA_antitoxin SocA_Panacea DUF2188 LRAT ? ? RteC T4SS-DNA_transf YWFCY

DATMGS010000018.1:63283-80548
Puia sp.

Pyr_redox_3 Pyr_redox_3 Peptidase_M28 Peptidase_M20 Rce1-like FKBP_C NamZ_C NamZ_N Formyl_trans_C Formyl_trans_N HTH_3 SocA_Panacea MqsR_toxin LysM LysM He_PIG Melibiase_2 Melibiase_C Glyco_hydro_123_N_2 GH123_cat Methyltr_RsmF_N Methyltr_RsmF_N Methyltr_RsmB-F Methyltranf_PUA ? Methyltransf_24 DHFR_1 Thymidylat_synt

DATMBO010000019.1:5441-15357
Bacillus sp. (in: firmicutes)

DUF2628 ParA ? ? SLH ? MqsA_antitoxin SocA_Panacea MqsR_toxin ? SLT_2 Peptidase_M23

DASRWC010000019.1:66587-12091
Flavobacteriaceae bacterium

DUF4140 DUF4139 Helicase_C DEAD DUF2807 PagL MqsA_antitoxin SocA_Panacea MqsR_toxin PALP OsmC Mur_ligase Mur_ligase_M Mur_ligase_C TPRs DUF4136 GTP1_OBG MMR_HSR1 ADK

WSYB01000008.1:80165-94469
Geobacteraceae bacterium

Complex1_30kDa Complex1_49kDa Complex1_49kDa Proton_antipo_M ParE_toxin Unstab_antitox ? HEPN_RiboL-PSP MqsA_antitoxin MqsR_toxin NTP_transf_8 DUF4342 Oxidored_q2 NADHdh Proton_antipo_N Proton_antipo_M Fer4_7 Oxidored_q6 Complex1_30kDa Complex1_49kDa Complex1_49kDa Proton_antipo_M

WSNY01000022.1:7759-17550
bacterium

LRR_14 LRR_14 LRRs SpecificRecomb ? MqsR_toxin SocA_Panacea ? ? ? DinB_2 Beta-lactamase2 2H ? MqsR_toxin MqsA_antitoxin SocA_Panacea ?

LDXJ01000061.1:231993-249281
Peptococcaceae bacterium CEB3

TPRs SEC-C Glyoxalase Cdd1 YutD-like Nitroreductase ? MqsA_antitoxin SocA_Panacea MqsR_toxin DUF5131 DEAD MZB DEAD Helicase_C ? Nitroreductase CW_binding_2 CW_binding_2 CW_binding_2 Flg_new Flg_new_2 Flg_new_2 Flg_new_2 Flg_new_2 Flg_new_2 Flg_new_2

JACNJH010000180.1:14801-8876
Candidatus Desulfatibia profunda

MqsR_toxin MqsA_antitoxin SocA_Panacea DUF6922 ? DDE_Tnp_1 DUF772 Succ_CoA_lig CoA_binding_2 ATP-grasp_5 DUF218 PagL

JACRTN010000001.1:67589-85379
Paraclostridium tenue

Enc34_ssDNA-bd HTH_26 HTH_3 DUF4003 PAS PAS PAS His-Kinase His-Kinase Zn_ribbon_15 MqsA_antitoxin SocA_Panacea MqsR_toxin CCDC113_CCDC96_CC Amidase_3 YvrJ SBP_bac_5 SBP_bac_5 ABC_membrane ABC_tran ABC_membrane ABC_tran FTP Peptidase_M4 Peptidase_M4_C

JAAVFS010000091.1:801-8476
Bacteroidales bacterium

Hydrolase_3 PRTase-CE ORC-CDC6-like ? PfkB MqsR_toxin ?

RKET01000103.1:1-6000
Prevotella sp.

MPI_cupin_dom MqsR_toxin MqsA_antitoxin SocA_Panacea DUF4435 ABC-ATPase RsgA_GTPase EamA EamA Lactate_perm

JACUUU010000330.1:1-3954
Anaerolineales bacterium

MqsA_antitoxin HicB ? Peptidase_S15 PepX_C Tc-R-P

JADBLR010000018.1:1-3551
Planctomycetota bacterium

? Methyltransf_23 PNPase MqsA_antitoxin SocA_Panacea ?

JADBKJ010000006.1:161232-179070
Thermoanaerobacter sp.

Cu_amine_oxidN1 DUF11 DUF11 DUF11 REase MFS_1 ? MFS_1 MFS_1 N6_N4_Mtase MqsR_toxin MqsA_antitoxin SocA_Panacea ? ? ? PemK_toxin Radical_SAM

SVGH01000001.1:68046-83655
Clostridiales bacterium

? WG WG WG WG ? ? ? ? HTH_3 SocA_Panacea MqsR_toxin AA_kinase ACT_9 Serpin NT5C NT5C YitT_membrane DUF2179 ABC_tran ABC_tran_Xtn ABC_tran ABC_tran_CTD Gate FeoB_C Gate FeoB_Cyto FeoB_N

SVGE01000005.1:107386-121300
Clostridiales bacterium

DUF421 YetF_N Mntp SLH SLH DUF1934 Abhydrolase_1 ? MqsA_antitoxin SocA_Panacea MqsR_toxin ADH_N ADH_zinc_N Mur_ligase_C Mur_ligase_M Dala_Dala_lig_C Dala_Dala_lig_N ? dCMP_cyt_deam_1 ? ?

SUYF01000018.1:378-11716
Prevotella sp.

? Acyl_transf_3 HTH_19 SocA_Panacea ? HTH_3 ? SMC_N MComp1 Trypsin CTD4 Lactamase_B DUF6602

JADFXJ010000002.1:16716-28112
Nitrospirae bacterium

Pirin Pirin_C ? ? DDE_5 ? FGS TIR MqsA_antitoxin MqsR_toxin ABC_tran PhoU PhoU DHquinase_I HD HD_5 cNMPBD ? Adenine_glyco

JACQOV010000061.1:32-9497
Deltaproteobacteria bacterium

MqsR_toxin HTH_3 SocA_Panacea Vgb_lyase Vgb_lyase Pyr_redox_dim Pyr_redox_2 2-oxoacid_dh E3_binding Biotin_lipoyl OxoGdeHyase_C Transket_pyr E1_dh 2-oxogl_dehyd_N YafQ_toxin PhdYeFM_antitox

JACRIR010000003.1:12587-27586
Candidatus Poribacteria bacterium

Amidohydro_2 DUF5615 DUF433 DUF3024 DUF4238 MazE_antitoxin MqsA_antitoxin SocA_Panacea MqsR_toxin AAA AAA_assoc_2 MgsA_C URO-D ? T2SSE GAF cNMPBD DUF218 HYDIN_VesB_CFA65-like_Ig Sortilin-Vps10 Sortilin-Vps10

JACRLB010000017.1:13465-31738
Chloroflexi bacterium

Virul_fac_BrkB HD_5 PAS Frankia_peptide DUF5317 ? BTAD TPR_MalT TPRs AAA_16 PHP DNA_pol3_alpha DNA_pol3_finger Intein_splicing HHH_6 N-OB MqsR_toxin MqsA_antitoxin SocA_Panacea Abhydrolase_6 DUF951 Glyco_transf_4 Glycos_transf_1 TsaE TsaD Acetylase

JAEINW010000126.1:9843-14878
Salinivirgaceae bacterium

AMP-binding AMP-binding_C_3 RuvC_III HNH_4 Cas9-BH Cas_Cas1 CRISPR_Cas2 MqsA_antitoxin SocA_Panacea ? HATPase_c_4 SLFN_AlbA_2

JADHWY010000050.1:97-7124
Dehalococcoidales bacterium

DUF4342 His_Phos_1 MqsR_toxin MqsA_antitoxin SocA_Panacea 5_3_exonuc_N 5_3_exonuc DNA_pol_A_exo1 DNA_pol_A Fapy_DNA_glyco H2TH zf-FPG_IleRS DUF4389 DUF4389 ?

VGYT01000103.1:6696-14163
Planctomycetota bacterium

AbiEi_4 ? Phage_portal_2 ? Resolvase Resolvase Trans_reg_C ? MqsA_antitoxin SocA_Panacea

VGSL01000491.1:618-1987
Deltaproteobacteria bacterium

MqsR_toxin MqsA_antitoxin SocA_Panacea

JAFGJO010000019.1:334-12950
Spirochaetales bacterium

DUF772 DDE_Tnp_1_6 ? ? HNH_2 ? MqsA_antitoxin SocA_Panacea MqsR_toxin PDDEXK_3 Acyltransferase Lon_C Lon_lid AAA URGCP LON_substr_bdg FHA GerE GDT1 GDT1 Receiver

JAFGIK010000043.1:354-5193
Candidatus Dojkabacteria bacterium

? ? MqsA_antitoxin SocA_Panacea MqsR_toxin ? TPRs ?

JAFGGR010000045.1:1-3626
Clostridiales bacterium

DJ-1_PfpI ? MqsA_antitoxin MqsR_toxin VIT1 VIT1 ? WG

JAFGFW010000139.1:16528-24022
Deltaproteobacteria bacterium

HTH_17 ResIII Helicase_C ? ? HTH_17 ? MqsA_antitoxin SocA_Panacea MqsR_toxin

JAFGEJ010000253.1:183-6781
Phycisphaerae bacterium

MqsA_antitoxin SocA_Panacea ? Thymidylat_synt ParA-Soj-NTPase ? KARI_C KARI_N

JAFGAA010000357.1:10558-10537
Anaerolineaceae bacterium

Acylphosphatase Form_Nir_trans ? MqsA_antitoxin SocA_Panacea ? NiFeSe_Hases NiFeSe_Hases Oxidored_q6 Fer2_4 NADH-G_4Fe-4S_3 Fer4_Nqo3 Fer4_7 Molybdop_Fe4S4 Molybdopterin Complex1_51K SLBB NADH_4Fe-4S Complex1_51K 2Fe-2S_thioredx HGTP_anticodon tRNA-synt_His

JAFGUF010000038.1:20803-31567
Clostridia bacterium

Inositol_P ? Cupin_2 HTH_3 ? DNA_alkylation ? HTH_20 DUF2087 MqsR_toxin MqsA_antitoxin SocA_Panacea Thia_YuaJ DUF2971 Peptidase_C39_2 ? ? ?

JAFLBJ010000001.1:573472-592678

Chitinophagales bacterium

2-ph_phosp ? GCV_T_C GCV_T Toast-rack_PspC-Cterm LiaI-LiaF-TM_PspC PspC PspC HAAS PadR 2OG-FeII_Oxy DIOX_N ? MqsR_toxin MqsA_antitoxin SocA_Panacea ExbD Thioredoxin_8 SusD_RagB SusD-like_3 Plug CarbopepD_reg_2 STN FecR_C FecR Sigma70_r4_2 Sigma70_r2

JAGAAH010000090.1:53-7320
Lachnospiraceae bacterium

HicA_toxin HicB_lk_antitox MqsR_toxin MqsA_antitoxin SocA_Panacea Sigma70_r2 EamA EamA Epimerase LysR_substrate HTH_1 Fer4_9 POR_N PFOR_II

JAGBQJ010000001.1:74858-88924
Bacteroidales bacterium

Amidase_2 HU-HIG DUF5675 ? ? ? HTH_3 MqsR_toxin ? SLFN_AlbA_2 HATPase_c_4 HTH_24 ASCH EVE N6_Mtase Methylase_S SMC_N ABC-ATPase Toprim Mrr_cat

JACDOK010000005.1:51042-64750
Peptococcaceae bacterium

Phage_integrase ? ? ParB_N HTH_ParB DUF192 ? Receiver CbiA MqsR_toxin MqsA_antitoxin SocA_Panacea Cu_amine_oxidN1 PQQ_2 PBECR3 ? T2SSE T2SSF

JAGBIO010000011.1:107537-126192
Clostridia bacterium

? Sigma70_r2 Sigma70_r4 ? Sigma70_r2 Sigma70_r3 Sigma70_r4 WG WG WG ? PHP DNA_pol3_alpha DNA_pol3_finger HHH_6 N-OB MqsR_toxin MqsA_antitoxin SocA_Panacea ? HsdM_N N6_Mtase Methylase_S Methylase_S DUF4145 REase ResIII Helicase_C EcoEI_R_C NUDIX ?

JAGCTT010000149.1:2558-20114
Treponema sp.

60KD_IMP PhdYeFM_antitox ? Glyphos_transf Aminotran_5 CTP_transf_like 60KD_IMP Phosphodiest MqsR_toxin MqsA_antitoxin SocA_Panacea Glyco_transf_4 Glycos_transf_1 Glyco_transf_4 Glycos_transf_1 AAA_14 Glyco_transf_4 Glycos_transf_1 Glycos_transf_2 ABC_tran

JAGNRN010000158.1:1677-4883
Fusobacteriaceae bacterium

Fic DUF6370 HTH_26 MqsR_toxin MqsA_antitoxin SocA_Panacea PRiA4_ORF3 ANCHR-like_BBOX DUF6567

JAFOFN010000256.1:889-2346
Treponema sp.

? MqsA_antitoxin MqsR_toxin

JAFQON010000007.1:65579-14399
Clostridia bacterium

HTH_7 HipA_C HATPase_c_3 HSDR_N_2 SecB MqsA_antitoxin MqsR_toxin ? ? Mur_ligase_M Mur_ligase_C ? ? Flg_new_2 ?

JAFPVR010000116.1:27982-42722
Bacteroidales bacterium

Cys_Met_Meta_PP Methyltransf_11 NQRA_N NQRA_2nd NQRA_SLBB Alpha_L_fucos PA14 NQR2_RnfD_RnfE FMN_bind Rnf-Nqr MqsR_toxin SocA_Panacea Rnf-Nqr Fer2 FAD_binding_6 NAD_binding_1 SRP54_N SRP54 SRP_SPB ? His_Phos_1 His_Phos_1 Asp_decarbox

JAFUKS010000178.1:100-3794
Lachnospiraceae bacterium

dTDP_sugar_isom MqsA_antitoxin SocA_Panacea MqsR_toxin Epimerase GDP_Man_Dehyd

JAFTWC010000050.1:70133-90302
Lachnospiraceae bacterium

? Cache_3-Cache_2 HAMP MCPsignal ? DUF1643 ? Macro VWA MqsR_toxin MqsA_antitoxin SocA_Panacea Glyco_hydro_10 Glyco_hydro_10 Abhydrolase_2 Peptidase_U32 ? ARG HTH_38

JAFTYK010000035.1:47476-56696
Ruminococcus sp.

? ? HTH_49 Resolvase DUF6809 ApeA_NTD1 HTH_3 DUF3990 MqsR_toxin MqsA_antitoxin SecB ? HTH_3 HTH_26 GntR ABC_tran

JAFSGI010000054.1:3443-14397
Ruminiclostridium sp.

ACT_8 AMP-binding_C_2 AMP-binding HlyIII DegV Thioredoxin Sigma70_r2 Sigma70_r4 AAA_14 DUF4143 MqsR_toxin MqsA_antitoxin SecB ParA-Soj-NTPase ParB_N HTH_ParB ?

JAFYLT010000202.1:1-9564
Oscillospiraceae bacterium

Clp_N Clp_N ? Acetylase His_Phos_1 SpoIVA_C SpoIVA_middle SpoIVA_ATPase SecB MqsA_antitoxin MqsR_toxin Amidohydro_3 Amidohydro_3 DUF1858 Methyltransf_18 DUF34_NIF3 Transcrip_reg TACO1_YebC_N

JAFXWY010000380.1:1-9257
Bacteroidales bacterium

HAD_2 HTH_3 ? CHASE3 ? Dynamin_N Zn_Ribbon_TF ? MqsR_toxin MqsA_antitoxin SocA_Panacea CPSase_L_D1

JAFZLA010000002.1:38133-53184
Treponema sp.

HTH_3 HipA_C Couple_hipA TPT EamA GumK_N Glyco_trans_1_4 UDPG_MGDP_dh_N UDPG_MGDP_dh UDPG_MGDP_dh_C MqsA_antitoxin SocA_Panacea MqsR_toxin Glycos_transf_2 GtrA_DPMS_TM Glycos_transf_2 ? Glycos_transf_2 Glycos_transf_1 DUF6492 ?

JAFZNG010000008.1:81544-104928
Paludibacteraceae bacterium

Fer2 FAD_binding_6 NAD_binding_1 SNF SNF 23S_rRNA_IVP CBS CBS ? Flg_new Flg_new ? MqsR_toxin MqsA_antitoxin Flg_new ? ? RMMBL Beta-Casp Lactamase_B Peptidase_S9 Acyl-ACP_TE Acyl-ACP_TE_C

JAFDZQ010000059.1:3464-14523
Bacteroidetes bacterium

? DUF3169 ? ? HTH_3 ? MqsA_antitoxin SocA_Panacea MqsR_toxin HTH_26 Cellulase CTP_synth_N GATase Acetylase CTP_transf_3 GatC Acyltransferase

JAFDVK010000028.1:1-10040
Acidobacteria bacterium

MM_CoA_mutase ? Prok-JAB ThiF Prok-E2_A ? Gly-zipper_Omp HNH DnaJ MqsR_toxin MqsA_antitoxin SocA_Panacea ? HNH_2 HTH_3 ? ?

JAHITX010000063.1:36914-54440
Proteobacteria bacterium

PEP-CTERM ? PBP_like_2 BPD_transp_1 BPD_transp_1 ABC_tran PhoU PhoU MqsR_toxin MqsA_antitoxin SocA_Panacea Methylase_S Methylase_S N6_Mtase 23S_rRNA_IVP ResIII HSDR_N_2 Porin_5 Porin_5 Porin_5 Lactamase_B STAS

JAFDJX010000084.1:700-4628
Deltaproteobacteria bacterium

MqsR_toxin MqsA_antitoxin SocA_Panacea ? ? TPP_enzyme_C TPP_enzyme_M TPP_enzyme_N

JAIBEP010000523.1:4666-11970
Youngiibacter sp.

CBS OmdA tRNA_edit adh_short_C2 HTH_1 LysR_substrate MR_MLE_C MqsA_antitoxin SocA_Panacea MqsR_toxin ? ?

JAHBUA010000023.1:1-10767
Cyclobacteriaceae bacterium

DUF998 DUF4293 HTH_19 SocA_Panacea MqsR_toxin DUF5930 DUF4230 GIY-YIG DPPIV_N Peptidase_S9 GIY-YIG ? ?

JAIBBK010000022.1:16796-38411
Ignavibacteria bacterium

PEP-utilizers_C PEP-utilizers PEP-utilisers_N Por_Secre_tail Methyltransf_12 Omp28 Por_Secre_tail Y1_Tnp Spectrin ? MqsR_toxin MqsA_antitoxin SocA_Panacea DNA_methylase HATPase_c_3 His-Kinase Receiver HsdM_N N6_Mtase Methylase_S Methylase_S PDDEXK_3

JAGQXI010000209.1:3928-10881
Lewinella sp.

Hydrolase_4 ? HTH_19 Peptidase_S24 TIM ? GAF_2 MqsA_antitoxin SocA_Panacea MqsR_toxin Pan_kinase

JAIOTF010000281.1:501-8043
Anaerolineaceae bacterium

Peptidase_A8 PseudoU_synth_2 S4 Aminotran_3 Beta_elim_lyase UPF0047 MqsA_antitoxin SocA_Panacea ? GPI 2-Hacid_dh_C Aldolase

JAIOSG010000105.1:9562-27018
Anaerolineaceae bacterium

? EamA EamA MFS_3 Dala_Dala_lig_C Met_asp_mut_E B12-binding Octopine_DH NAD_Gly3P_dh_N MqsA_antitoxin SocA_Panacea MqsR_toxin PRK RE_EcoO109I N6_N4_Mtase Cupin_2 AIRS_C AIRS TPRs TPRs TPRs TPRs TPRs His-Kinase HisKA_3 GAF_2 GAF_2 GAF_2 PAS PAS

JAJQCA010000032.1:6403-12752
Alistipes sp.

OMP_b-brl ? MqsA_antitoxin SocA_Panacea IMS_C IMS_HHH_1 IMS Calycin_like LplA-B_cat Aldolase_II Aldolase_II ADH_zinc_N ADH_N adh_short Aldolase_II

JAJPZD010000242.1:1-6645
Prevotella sp.

HKD MqsR_toxin MqsA_antitoxin SocA_Panacea PhdYeFM_antitox RelE Lcl_C LRRs LRRs LRRs DUF3791 DUF3990

JAJFUK010000128.1:61883-73965
Actinomycetes bacterium

T5orf172 ? DUF3800 ? ? ? ? MqsR_toxin MqsA_antitoxin SocA_Panacea SoFic-like_C ? DUF2680 His-Kinase His-Kinase HAMP Trans_reg_C Receiver IMPDH

JABUSH010000003.1:150252-163163
Erysipelotrichaceae bacterium

Flavin_Reduct ATP_bind_3 Peptidase_M19 DNA_binding_1 DAHP_synth_1 TM1586_NiRdase MqsA_antitoxin SocA_Panacea MqsR_toxin Chromate_transp Chromate_transp MgtE CBS CBS MgtE_N MgtE_N MgtE CBS MatE MatE

JABUUK010000061.1:12165-24924
Muribaculaceae bacterium

Radical_SAM Radical_SAM_C Omp28 PAS MBOAT HlyD_D23 OEP ACR_tran MqsR_toxin

JAIPEV010000081.1:722-7372
Halanaerobiales bacterium

TP_methylase tRNA_bind Anticodon_1 tRNA-synt_1g tRNA-synt_1g CoA_binding Put_DNA-bind_N Lactamase_B_2 SPOUT_MTase ? MqsA_antitoxin ?

JAFLTT010000006.1:48909-60326
Muribaculaceae bacterium

MFS_1 ? TPRs FMN_bind Fer4_5 Fer4_5 PHP DUF418 MqsA_antitoxin ? DDE_Tnp_IS66 DDE_Tnp_IS66_C TnpB_IS66 ? ORF6N ? ? Peptidase_S24

JADFMJ010000022.1:4074-20238
Nitrospinota bacterium

PhosphMutase ? ? NucS_C ? Peptidase_M29 MqsA_antitoxin VESA1_N MqsR_toxin DUF2283 DNA_processg_A Lysine_decarbox Aldedh MacB_PCD FtsX MacB_PCD FtsX ABC_tran HlyD_D23

JAGMCV010000126.1:1-3954
Candidatus Atribacteria bacterium

MqsA_antitoxin SocA_Panacea MqsR_toxin ? ? NamZ_C NamZ_N ?

JAGMBE010000414.1:290-1885
Candidatus Aerophobetes bacterium

? MqsA_antitoxin SocA_Panacea MqsR_toxin

JAGLZP010000053.1:1-3812
Spirochaetaceae bacterium

N6_Mtase Methylase_S DUF2188 MqsR_toxin MqsA_antitoxin SocA_Panacea

JAGLWZ010000056.1:1-5700
Bacteroidales bacterium

? ? B12-binding Radical_SAM ? ? MqsA_antitoxin SocA_Panacea

JAGLVY010006978.1:273-1216
bacterium

MqsR_toxin MqsA_antitoxin

JAGLTS010000076.1:29523-44038
Phycisphaerales bacterium

N6_N4_Mtase ParB_N Pox_D5 DUF5906 D5_N ? HTH_23 Sigma70_r2 ? Rrf2 SAP MqsR_toxin MqsA_antitoxin SocA_Panacea ? ? FH2 Zn_ribbon_recom Recombinase Resolvase DUF2924 Ribosomal_L31 Glyoxalase

JAGLQV010001082.1:446-2561
Deltaproteobacteria bacterium

HicA_toxin UPF0150 MqsA_antitoxin SocA_Panacea MqsR_toxin

JAGLOJ010000039.1:287-6024
Candidatus Sabulitectum sp.

? MqsA_antitoxin SocA_Panacea SLFN_AlbA_2 HATPase_c_4 Fic-like_C DUF4276 Ppnp PmbA_TldD_1st PmbA_TldD_3rd Calcineurin

JAGLNS010000015.1:74-1374
Spirochaetales bacterium

MqsR_toxin MqsA_antitoxin SocA_Panacea

JAGLHB010000874.1:1-2600
Candidatus Paceibacterota bacterium

DUF643 MqsR_toxin MqsA_antitoxin SocA_Panacea

JALNWI010000303.1:2376-2108
Candidatus Muiribacterium halophilum

? MqsA_antitoxin SocA_Panacea MqsR_toxin OmpH

JALOAY010000308.1:1-3334
Limnochordia bacterium

DUF2399 URO-D MqsA_antitoxin SocA_Panacea MqsR_toxin

JALPYB010000025.1:217-10973
Peptococcaceae bacterium

Orn_Arg_deC_N DHH DHHA1 CBS CBS PolyA_pol PolyA_pol_RNAbd tRNA_NucTran2_2 Peptidase_M50 Peptidase_M50 tRNA-synt_1b ? HTH_31 HTH_Crp_2 cNMPBD Fer4_21 Fer4_7 Hydrolase_3 PELOTA_1 StiP TRSP

JAMCSN010000214.1:1520-10969
Bacteroidetes bacterium

Suc_Fer-like Aldo_ket_red DUF3410 2-Hacid_dh_C Band-7 NfeD NfeD_membrane NfeD1b_N DoxX DUF2326 MqsR_toxin MqsA_antitoxin SocA_Panacea N6_Mtase

JAMPHQ010000002.1:102365-121146
Eubacterium sp.

Seryl_tRNA_N tRNA-synt_2b DUF3592 DUF3881 MCPsignal HAMP CZB ABC_tran ABC2_membrane_6 ABC2_membrane_6 MqsR_toxin MqsA_antitoxin SocA_Panacea Peptidase_S51 ? PAS His-Kinase His-Kinase Receiver GxDLY Lipase_GDSL_3 His-Kinase His-Kinase Receiver Receiver Glycophorin_A 5_nucleotid_C Calcineurin

JAMPGP010000616.1:1-3191
Muribaculum sp.

? HTH_19 SocA_Panacea ? PcfJ

RJUD01000006.1:143771-160481
Lacihabitans sp. CS3-21

FAA_hydrolase NUDIX AAA_14 DUF4143 LigB DUF2461 ? MqsA_antitoxin SocA_Panacea ? Radical_SAM ? Glyco_hydro_67C Glyco_hydro_67M Glyco_hydro_67N Esterase LiaF-TM FAD_binding_4 FAD-oxidase_C Fer4_17 ?

JAKSKR010000005.1:28263-44632
Paludibacteraceae bacterium

PP-binding ? ? Glyco_hydro_108 PG_binding_3 Serpin Glyco_hydro_39 CBM_6 Por_Secre_tail SocA_Panacea ? AAA-ATPase_like PDDEXK_9 Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 TIR TIR TIR TIR Lcl_C Slo-like_RCK RyR

JANLFY010000010.1:94502-123565
Bacillota bacterium

BPD_transp_1 BPD_transp_1 G3P_antiterm Na_sulph_symp Hydrolase_6 Hydrolase_like DAO Fer2_BFD Pyr_redox_2 DUF1667 MqsR_toxin MqsA_antitoxin SocA_Panacea SNF2-rel_dom Helicase_C MmeI_Mtase Eco57I Eco57I DEAD DEAD Helicase_C MZB ? HTH_IclR IclR_C DctP

JANLFS010000135.1:4993-6265
Bacillota bacterium

BCA_ABC_TP_C ABC_tran BCA_ABC_TP_C ABC_tran HTH_31 SocA_Panacea MqsR_toxin HTH_21 rve HTH_28 HTH_Tnp_1 HTH_Tnp_1 Trm56 HTH_50 AAA_lid_14 NtrC-AAA PAS

JAILDZ010000053.1:85-2640
Lachnospiraceae bacterium

ResIII HSDR_N_2 ? ? ? MqsR_toxin

JAPHSS010000024.1:12480-23059
Gammaproteobacteria bacterium

EcoRII-C HTH_23 DNA_methylase Vsr GIY-YIG Sel1 Sel1 DUF6164 Zn_peptidase_2 MqsR_toxin MqsA_antitoxin SocA_Panacea ? ? ? ? ?

JAPLKL010000023.1:7365-24009
Elusimicrobia bacterium

Fer4_7 Glycos_transf_2 SRAP GIY-YIG EcoEI_R_C Helicase_C ResIII Methylase_S Methylase_S N6_Mtase HsdM_N ? MqsA_antitoxin SocA_Panacea ? ? ? ? ? ?

JAPLJT010000136.1:1-3826
Deltaproteobacteria bacterium

? ? ParB_N HTH_ParB MqsR_toxin MqsA_antitoxin SocA_Panacea

JAPLHM010000239.1:1-4531
Chloroflexota bacterium

DAO ? MqsA_antitoxin SocA_Panacea TauE CCG CCG LUD_dom

JAPLGQ010000062.1:5727-10599
Chloroflexota bacterium

Methylase_S REase ResIII Helicase_C EcoEI_R_C ? MqsA_antitoxin SocA_Panacea Phage_integrase TIM UPRTase RuvB_N AAA_lid_4 RuvB_C Queuosine_synth UDG

JAPLDV010000839.1:116-1520
Bacteroidia bacterium

MqsR_toxin MqsA_antitoxin SocA_Panacea

JAPLDR010000039.1:10502-26567
Bacteroidia bacterium

LTD LTD CHU_C Semialdhyde_dhC Semialdhyde_dh GTP-bdg_N GTP-bdg_M MMR_HSR1 TonB_C Fib_succ_major ? MqsA_antitoxin SocA_Panacea ? ? ? DUF6261 ? Chal_sti_synt_C Chal_sti_synt_N ICMT Cation_ATPase_N E1-E2_ATPase Hydrolase Cation_ATPase_C

JAPKYR010000185.1:1-11317
bacterium

? ATP19 DUF6132 Lyase_aromatic PepX_C Peptidase_S15 MqsA_antitoxin MqsR_toxin AAA_3 AAA_lid_2 DUF5074 LVIVD LVIVD LVIVD LVIVD Peptidase_S41 PDZ_Tricorn Tricorn_C1 PD40 PD40 PD40

JAPNUW010000504.1:1-4163
Bacillota bacterium

? HVO_1552_C OrfB_IS605 MqsR_toxin MqsA_antitoxin SocA_Panacea DUF6745 ?

JAPUKP010000161.1:1-10447
Proteobacteria bacterium

? MqsA_antitoxin SocA_Panacea MqsR_toxin ? HATPase_c_3 Radical_SAM DndB ? SNF2-rel_dom Helicase_C ?

JAQIBN010000263.1:9593-21351
Spirochaetia bacterium

PfkB LacI Peripla_BP_3 GFO_IDH_MocA_C3 GFO_IDH_MocA Acetylase rve_3 ? HTH_3 MqsR_toxin MqsA_antitoxin SocA_Panacea ? Asp_Glu_race HIT 2-Hacid_dh_C Amidohydro_1 AP_endonuc_2

JAQKAW010000003.1:6040-27500
Tenacibaculum dicentrarchi

? ? ? T1RH-like_C T1R_D3-like SWI2_SNF2 REase PDDEXK_4 Methylase_S Methylase_S N6_Mtase HsdM_N MqsR_toxin MqsA_antitoxin SocA_Panacea HTH_3 HipA_C AAA_lid_3 AAA STY-kinase KilA-N HSP70 ?

JAQUHE010000180.1:499-3384
Atribacterota bacterium

MqsR_toxin MqsA_antitoxin SocA_Panacea ArAE_1

JAQUGL010000126.1:233-8953
Bacteroidales bacterium

MqsA_antitoxin SocA_Panacea ? REase SWI2_SNF2 T1R_D3-like ? Methylase_S Methylase_S Virulence_RhuM Fic HsdM_N N6_Mtase

JAQUCV010000001.1:82018-97903
Syntrophomonadaceae bacterium

Ribosomal_S18 MazG-like DUF2232 Ribosomal_L9_N Ribosomal_L9_C NtrC-AAA Lon_C Adenylsucc_synt Fic HTH_DeoR MqsR_toxin HTH_3 SocA_Panacea SCIFF Radical_SAM SPASM LysM G5 Peptidase_M23 Pyr_redox_2 Peptidase_MA_2 EPSP_synthase

JAQTZZ010000032.1:35120-45955
Candidatus Wallbacteria bacterium

NifU_N DUF2157 GDYXXLXY ANKs Nitroreductase Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 ? ? MqsR_toxin MqsA_antitoxin SocA_Panacea ? ? Sel1 Sel1 Sel1 Sel1 ORF6N

JAQWCA010000279.1:1-3109
Tepidanaerobacteraceae bacterium

DUF4351 EIID-AGA MqsR_toxin MqsA_antitoxin Yae1_N DUF4351

JAQYLV010000028.1:1-3514
Solobacterium sp.

? MqsA_antitoxin SocA_Panacea MqsR_toxin Isochorismatase NAPRTase NAPRTase_N

JAIDMJ010000574.1:6-7783
Prevotella sp.

OMP_b-brl ? OMP_b-brl_2 MqsR_toxin SocA_Panacea PriA_3primeBD DEAD Zn_ribbon_PriA Helicase_C PriA_C ?

JARBUC010000045.1:392-12290
Bacillales bacterium

Ribosomal_S9 Phage_integrase Phage_int_SAM_3 Arm-DNA-bind_4 ? HTH_Tnp_ISL3 ? ? ? MqsR_toxin MqsA_antitoxin SocA_Panacea HSDR_N_2 N6_Mtase Methylase_S ?

JAOUKI010000039.1:1-6072
Dehalococcoidia bacterium

NADH_4Fe-4S SLBB Complex1_51K 2Fe-2S_thioredx 2Fe-2S_thioredx Fapy_DNA_glyco H2TH zf-FPG_IleRS MqsR_toxin MqsA_antitoxin SocA_Panacea ? DUF4389 DUF4389

JASEIX010000031.1:285-6625
Anaerolineales bacterium

? MqsA_antitoxin SocA_Panacea DUF2188 HTH_32 N6_N4_Mtase Transketolase_N Transket_pyr Transketolase_C FGGY_N FGGY_C

JASGVV010000058.1:246-6594
Bacillota bacterium

DUF1706 ArsC MqsR_toxin MqsA_antitoxin SocA_Panacea ODP Flavodoxin_1 Nitroreductase ? Hydrolase_4 BD-FAE

JASGTE010000104.1:640-3648
Atribacterota bacterium

MqsR_toxin MqsA_antitoxin SocA_Panacea ArAE_1

JASGTE010000118.1:261-2503
Atribacterota bacterium

Flg_new MqsA_antitoxin SocA_Panacea MqsR_toxin

JAJSZQ010000067.1:2837-12426
Deltaproteobacteria bacterium

? ? ? ? PIN DUF2281 RE_endonuc ResIII MqsR_toxin MqsA_antitoxin SocA_Panacea ?

JAJSZX010000150.1:3-7402
Deltaproteobacteria bacterium

? PIN HTH_37 Gp49 DUF4131 Competence Lactamase_B MqsR_toxin MqsA_antitoxin SocA_Panacea

JAURZL010000009.1:39985-55018
Anaerolineaceae bacterium

LacI Peripla_BP_3 ? PD40 PD40 ? S-methyl_trans ? cobW Intu_longin_3 MqsA_antitoxin SocA_Panacea DUF1638 RACo_C_ter Raco_middle RACo_linker Fer2 B12-binding_2 B12-binding Glyco_hydro127C Glyco_hydro127M Beta-AFase-like_GH127_cat Lip_prot_lig_C LplA-B_cat PSCyt1

JAUXUL010000048.1:10710-24233
Anaerolineales bacterium

HSP20 ? ? SIMPL ? BrnT_toxin CopG_antitoxin MqsR_toxin MqsA_antitoxin SocA_Panacea ? YhhN HATPase_c_3 ? PMT_2 DMT_YdcZ

JAUYMS010000003.1:141054-151498
Candidatus Omnitrophota bacterium

DHH DHHA1 TruB_N TruB_C_2 FAD_syn Flavokinase Ribosomal_S15 RNase_PH RNase_PH_C PNPase RNase_PH RNase_PH_C KH_1 S1 dUTPase FtsK_4TM FtsK_alpha FtsK_SpoIIIE FtsK_gamma DUF3576 MqsA_antitoxin SocA_Panacea HTH_Tnp_1

JAISEO010000020.1:12439-35971
Coriobacteriales bacterium

Flavokinase FAD_syn ClpB_D2-small MerR_1 DnaJ_C DnaJ GrpE HSP70 ? MqsR_toxin MqsA_antitoxin SocA_Panacea DUF3320 MTES_1575 AAA_12 AAA_11 AAA_11 DUF4011 DUF2510 BipA_C EFG_C GTP_EFTU_D2 GTP_EFTU DUF4143 AAA_14 MazG MazG PemK_toxin

JAITAX010000224.1:1-7914
Eubacteriales Family XIII. Incertae Sedis bacterium

MqsR_toxin MqsA_antitoxin ? UVR UvrB Helicase_C UvrB_inter ResIII Glyoxalase GAF PAS NtrC-AAA AAA_lid_14 Amidohydro_1

JAVEWN010000011.1:7987-22295
Anaerolineales bacterium

SpoIIM RDD Esterase Zn_peptidase_2 Peptidase_S41 PDZ_6 CtpB_N-like 4HBT_2 4HBT_2 MmeI_C MmeI_TRD MmeI_Mtase MmeI_hel MmeI_N MqsR_toxin MqsA_antitoxin SocA_Panacea SET Isochorismatase VanZ RNase_H Cauli_VI ? Sigma70_r4_2 Sigma70_r2

JAXDVY010000131.1:70-7251
Sodaliphilus sp.

MqsR_toxin SocA_Panacea GFO_IDH_MocA_C3 GFO_IDH_MocA DUF4359 GYF LytTR Receiver His_kinase FtsX MacB_PCD ABC_tran

JAXHVG010000026.1:160-7217
candidate division WOR-3 bacterium

Fic ParE_toxin PhdYeFM_antitox MqsA_antitoxin SocA_Panacea MqsR_toxin LVIVD LVIVD LVIVD LVIVD LVIVD LVIVD LVIVD LVIVD LVIVD LVIVD LVIVD LVIVD Por_Secre_tail

JAXHMA010000004.1:609947-623654
Balneolaceae bacterium

? ? ? Transglutaminase ? ? ? ? MqsA_antitoxin SocA_Panacea ? Prok-JAB ThiF AbiEii DUF6088 UvrD-helicase

JAYENW010000030.1:1-12829
Chloroflexota bacterium

YgjP-like DUF7376 PDDEXK_1 HAD WDR55 GTP_EFTU EF-G_D2 EFG_III EFG_C BipA_C MqsR_toxin MqsA_antitoxin SocA_Panacea DUF5615 DUF433 Terminase_6C SPP1_portal PhdYeFM_antitox GGDEF Receiver

JAYQWI010000178.1:1-4705
Candidatus Parabeggiatoa sp.

TPRs STY-kinase LRRs MqsR_toxin MqsA_antitoxin ? ? ?

JAXJWM010000008.1:1-11712
Paludibacteraceae bacterium

N6_Mtase Methylase_S Methylase_S ? REase SWI2_SNF2 T1R_D3-like MqsR_toxin MqsA_antitoxin SocA_Panacea LPD3 Rnf-Nqr Rnf-Nqr FMN_bind NQR2_RnfD_RnfE Fer4_9

JAXJUP010000006.1:1-1973
Paludibacteraceae bacterium

CusB_dom_1 HlyD_3 MqsA_antitoxin ?

JAACTU010000056.1:4075-16524
Nitrospirae bacterium

OmpA PD40 PD40 PD40 TolB_N TonB_2 ExbD MotA_ExbB DUF1848 PNPOx_N ? MqsA_antitoxin SecB DUF6933 ORF6N Ribosomal_S7e Methylase_S Methylase_S DUF559 DUF433

JAAYZT010000004.1:163-2184
Dehalococcoidia bacterium

MqsR_toxin PIG-L

JAAYIF010000032.1:96805-110083
Clostridia bacterium

NUDIX ? SpoIIM STE2 Phage_int_SAM_1 Phage_integrase Metalloenzyme Glycos_trans_3N Glycos_transf_3 PYNP_C ? MqsA_antitoxin SocA_Panacea Peptidase_S11 PBP5_C STAS His-Kinase Sigma70_r2 Sigma70_r3 Sigma70_r4 SpoVAC_SpoVAEB SpoVAD

JAAZLW010000234.1:200-1703
Bacillota bacterium

? ? MqsA_antitoxin SocA_Panacea

JAAZLY010000156.1:1-6448
Bacillota bacterium

? ? MqsA_antitoxin SocA_Panacea ? DUF6591 Big ? His-Kinase Fer4_10 Fe_hyd_lg_C FeS

JAAZNE010000003.1:35624-51715
Chloroflexota bacterium

PDZ_2 Trypsin Zn_ribbon_3 baeRF_family10 TPRs PTP-SAK Metallophos_2 Acetylase MqsA_antitoxin SocA_Panacea ? ? Radical_SAM ? LytR_cpsA_psr Maf Transpeptidase PBP_dimer MecA_N RsgA_GTPase RsgA_N

JAAZON010000639.1:1-4533
SAR324 cluster bacterium

? DUF2089 AbiEi_4 ? MqsR_toxin MqsA_antitoxin SocA_Panacea

WTBT01000107.1:28955-6329
Candidatus Aegiribacteria sp.

? DUF1451 ? ? MqsA_antitoxin SocA_Panacea AAA_15 ? ? Peptidase_C39_2 ? Por_Secre_tail Peptidase_C25

JAADGE010000078.1:11768-29856
Chlorobiota bacterium

ABC-ATPase RloB Methylase_S Methylase_S ? N6_Mtase HsdM_N COR-B COR-A AP-GTPase LRR_14 MqsA_antitoxin SocA_Panacea ? Cnn_1N HTH_17 DUF5906 DUF3854 ? ? Arm-DNA-bind_5 Phage_int_SAM_5 Phage_integrase AA_kinase

JABLXA010000033.1:16038-17992
Bacillota bacterium

Bac_transf ? Peptidase_M78 HTH_31 MZB Helicase_C DEAD ? ? ? MqsR_toxin HTH_31 SocA_Panacea rve_2 ? HicB UPF0150 Hemerythrin MazE_antitoxin SocA_Panacea MqsA_antitoxin

JABMQH010000057.1:1108-5218
Candidatus Omnitrophota bacterium

HTH_17 MqsR_toxin MqsA_antitoxin SocA_Panacea Trypsin Methyltransf_31 Smr

JAAXTO010000814.1:1-1664
Clostridiales bacterium

? HTH_3 SocA_Panacea MqsR_toxin

MGUL01000186.1:5717-22535

Elusimicrobia bacterium RBG_16_66_12

PadR Fer4_5 Fer4_5 PHP ? ? Phage_int_SAM_1 Phage_integrase ? MqsA_antitoxin SocA_Panacea ? Calcineurin RNase_T Calcineurin HTH_3 Couple_hipA SNase

MGXK01000139.1:7828-13831

Gammaproteobacteria bacterium RBG_16_51_14

? NfeD1b_N NfeD_membrane NfeD Band-7 DUF6946 ? MqsA_antitoxin SocA_Panacea TANGO2 Radical_SAM ? ? ? STY-kinase TPRs Nitroreductase Nitroreductase

MDZB01000141.1:3470-15571
Hymenobacter lapidarius

? Resolvase DUF6602 ThiF Prok-JAB SMODS-polBeta DUF2188 MqsA_antitoxin SocA_Panacea MmeI_N MmeI_hel MmeI_Mtase MmeI_TRD Dimerisation HTH_3 ? MqsR_toxin

MVQQ01000069.1:165-5374
Syntrophorhabdus sp. PtaB.Bin184

? Polbeta HicB HTH_3 Couple_hipA HipA_C ? MqsR_toxin MqsA_antitoxin SocA_Panacea

NOXE01000027.1:3909-13623
Priestia aryabhattai

ResIII Helicase_C Peptidase_S78 Phage_capsid ? Terminase_4 ? RE_NgoPII DNA_methylase ? MqsA_antitoxin SocA_Panacea

NTWK01000031.1:94467-107581
Bacillus cereus

Thioredoxin_4 Transgly NUDIX Transcrip_reg TACO1_YebC_N SMI1_KNR4 Imm11 PriX ? MqsA_antitoxin SocA_Panacea ? Terminase_4 ? Phage_connect_1 Phage_tail_2 Phage_capsid Peptidase_S78

PCTQ01000156.1:6072-17310
Elusimicrobia bacterium CG22_combo_CG10-13_8_21_14_all_63_91

? MacB_PCD HlyD_3 ABC_tran OEP OEP MqsA_antitoxin SocA_Panacea ? ? PRiA4_ORF3 Phage_integrase NTP_transf_2 zf-IS66 DDE_Tnp_IS66 DUF1016_N YhcG_C Phage_integrase

PHBZ01000063.1:14867-22103
Chloroflexi bacterium HGW-Chloroflexi-10

Fer4_6 Fer4 ? GAF_2 HD_5 DUF6273 MqsA_antitoxin SocA_Panacea ? ? HigB_toxin

QSLN01000015.1:23655-12348
Ammonifex thiophilus

? AAA_10 MqsA_antitoxin SocA_Panacea MqsR_toxin T2SSE ? TcpE RRXRR HNH TrbC bPH_2 TpcC

QZIE01000073.1:1-12941
Candidatus Parcubacteria bacterium

D5_N Toprim_2 ? MqsR_toxin MqsA_antitoxin SocA_Panacea ? ? ParB_N SoPB_HTH ? Peptidase_M78 HTH_3 EcoEI_R_C Helicase_C ResIII REase DUF4145

QNAI01000183.1:23-7443
Verrucomicrobia bacterium

Hydrolase_4 HTH_8 AAA_lid_14 NtrC-AAA Receiver His-Kinase His-Kinase GAF_3 ? MqsA_antitoxin SocA_Panacea HTH_28

QNET01000011.1:11814-27906
Candidatus Parabeggiatoa sp. nov. 3

Amidoligase_2 ? OEP OEP PCMT ThiC-associated ThiC_Rad_SAM ? ? MqsR_toxin HTH_31 ? iSTAND SocA_Panacea ? MoxR-AAA LysM

RPPU01000087.1:1-11214
Desulfobacteraceae bacterium

FtsX MacB_PCD Q_salvage HAD_2 MqsA_antitoxin SocA_Panacea MqsR_toxin DUF1848 AbiEii ? ? AbiEi_4 AbiEi_1 ABC_tran TOBE_2 BPD_transp_1

SLTJ01000032.1:139978-155436
Clostridia bacterium

T2SSE T2SSF Flp_Fap Peptidase_A24 ? ? ? MqsR_toxin MqsA_antitoxin Cu_amine_oxidN1 PKD PKD PKD_4 PKD_4 ? ? RadC DUF6075

SOKM01000030.1:123-1507
Candidatus Cloacimonadota bacterium

MqsA_antitoxin SocA_Panacea ?

QYOV01000103.1:744-8273
Dehalococcoidia bacterium

PAS HisKA_3 His-Kinase Receiver GerE Frankia_peptide AMP-binding_C AMP-binding ACAS_N ? ? HTH_3 SocA_Panacea MqsR_toxin

QYOT01000072.1:1-5090
bacterium

Eco57I MmeI_Mtase ? MqsA_antitoxin SocA_Panacea

CP107201.1:410237-425426
Oscillospiraceae bacterium

His-Kinase His-Kinase Receiver Trans_reg_C HTH_3 MobC NikA-like Relaxase SecB MqsA_antitoxin MqsR_toxin TTRAP ArdA ArdcN Peptidase_M78 LPD16 LPD25 YodL DUF4316 Toprim Topoisom_bac Toprim_Crpt CD1107-like DUF4315 NLPC_P60 ?

NZ_AEDV01000024.1:3910-15361
Streptococcus mitis SK597

Pribosyltran_N Pribosyl_synth PDDEXK_2 PDDEXK_2 PrsW-protease MqsA_antitoxin SocA_Panacea His_Phos_1 MqsR_toxin Pro_CA Zn_ribbon_LapB AAA_25 ChlI TnpB_IS66

NC_014171.1:945619-961825
Bacillus thuringiensis BMB171

DnaB_C DnaB AIPR ? MqsR_toxin HTH_31 SocA_Panacea DnaB_2 TauE TrmB Regulator_TrmB nSTAND3 Mrr_cat PRA-PH ? ? ? Acetyltransf_4 ?

NZ_JAJHUI010000004.1:171049-178404
Bacillus subtilis

YjcQ PBSX_XtrA ? ? ? HTH_28 ? ? DUF1064 ? MqsR_toxin MqsA_antitoxin ? ? Phage_terminase

NZ_JADPEQ010000004.1:2749-19762
Clostridium perfringens

SmpB Nucleos_tra2_N Gate Nucleos_tra2_C Pribosyltran SecG ? ? GntR TPRs TPRs TPRs OB_RNB CSD2 RNB S1 ? MqsA_antitoxin SocA_Panacea MqsR_toxin N6_Mtase DUF1097 Peptidase_S41 Peptidase_S41

NZ_CP009576.1:159770-170491
Listeria ivanovii subsp. londoniensis

? ? HALZ ? Phage_holin_1 Peptidase_M15_4 PSA_CBD PSA_CBD ? ? Peptidase_M10 Internalin_N LRRs LRR_adjacent GW Internalin_N LRRs LRRs LRR_adjacent ? MqsA_antitoxin Phage_XkdX FHIPEP

NZ_CP009576.1:780988-795189
Listeria ivanovii subsp. londoniensis

HTH_19 ? ? PhageMin_Tail ? Prophage_tail Peptidase_S74 BppU_N PPP1R21_C Phage_holin_1 Peptidase_M15_4 PSA_CBD PSA_CBD ? DUF3267 ? MqsA_antitoxin Phage_integrase

NZ_BJOD01000016.1:57708-78512
Brevibacillus agri

? MqsA_antitoxin SocA_Panacea MqsR_toxin ? Transgly_assoc ANKs ANKs FTP PepSY Peptidase_M4 Peptidase_M4_C SBP_bac_5 ? CN_hydrolase SDF ABG_transport Peptidase_M20 ? Esterase

NZ_CABKTG010000019.1:45563-61112
Faecalispora jeddahensis

SIS SIS GATase_6 ? ? ? Phage_TAC_5 Sigma70_r2 Sigma70_r4 SecB FRG GMP_synt_C NAD_synthase GATase Adenylsucc_synt ADSL_C Lyase_1 ? MqsR_toxin MqsA_antitoxin DUF3862 Zn_Ribbon_1

NZ_CP074441.1:1250389-1265230
Weissella ceti

Ribosomal_L32p NLPC_P60 LysM CHAP AAA AAA_lid_9 AAA_2 ClpB_D2-small TPP_enzyme_C TPP_enzyme_M TPP_enzyme_N AAL_decarboxy HTH_3 MqsR_toxin Usp PTS_EIIC_2 2-Hacid_dh_C CLP_protease WhiA_N LAGLIDADG_WhiA HTH_WhiA ADH_N ADH_zinc_N MarR_2

NZ_PNFF01000002.1:255998-283936
Bacillus methanolicus

DUF2726 ? DUF3298 PdaC ? ? Beta-lactamase2 HTH_17 Sigma70_r4_2 MqsA_antitoxin SocA_Panacea MqsR_toxin DUF6359 CHB_HEX_C_1 DUF5689 CHB_HEX_C_1 Synaptojanin SbsC_C Calcineurin 5_nucleotid_C ? IstB_IS21 PGA_cap RVT_1 GIIM

NC_015519.1:2450520-2466850
Tepidanaerobacter acetatoxydans Re1

SBP_bac_3 PilX_N SNF SNF Beta_elim_lyase PAS NtrC-AAA AAA_lid_14 HTH_8 Cu_amine_oxidN1 HTH_3 MqsR_toxin Acyl-ACP_TE_C Acyl-ACP_TE FCD GntR Lactate_perm ETF ETF_alpha ETF FAD-oxidase_C FAD_binding_4 Linocin_M18

NZ_CM122994.1:4012115-4018497
Bacillus subtilis

? HTH_17 PBSX_XtrA Sigma70_r4_2 DUF1064 ? DUF7128 ? ? ? ? ? DUF3792 MqsA_antitoxin ?

NZ_AXDC01000010.1:31127-40885
Caldanaerobacter subterraneus subsp. yonseiensis KB-1

T2SSF ? TadE ? ParA-Soj-NTPase ? ? MqsR_toxin MqsA_antitoxin ? ? Peptidase_A24 Transposase_31 HTH_7 ? ?

NZ_ATUV01000001.1:245454-260852
Hippea alviniae EP5-r

? AbiEii PIN ? ? TrwB_AAD_bind TrwC AAA_30 SH3_13 Viral_helicase1 ? MqsA_antitoxin SocA_Panacea BPD_transp_2 ? ? ? ? ?

NZ_AUFP01000014.1:30810-43061
Segatella albensis DSM 11370 = JCM 12258

Glycos_transf_2 EpsG Glycos_transf_2 DUF4422 NAD_binding_8 GLF Epimerase MqsR_toxin HTH_19 SocA_Panacea SLBB SLBB SLBB SLBB_2 SLBB SLBB Poly_export GNVR Wzz Glyco_transf_4 Glycos_transf_1

NZ_JAGM01000061.1:1-10362
Heyndrickxia ginsengihumi J57

HTH_Tnp_1 ? ? MqsA_antitoxin SocA_Panacea S4 Ribosomal_S4 GAF_2 GAF GGDEF GAF_2 TetR_N Aminotran_5 EzrA

NZ_KE384204.1:160242-174090
Butyrivibrio sp. FCS006

ABC_tran AAA_13 FtsX_ECD FtsX ? MqsA_antitoxin SocA_Panacea ? AAA_15 ABC-ATPase RloB DUF3450 CC_PcsB Peptidase_M23 ? PP2C_2 STY-kinase MqsR_toxin CtpB_N-like PDZ_6 Peptidase_S41 Glyco_transf_4 Glycos_transf_1 Resolvase

NZ_CP048273.1:1701128-1713106
Bacillus sp. NSP9.1

? ? ? XhlA Holin_SPP1 Glyco_hydro_25 LysM LysM MqsR_toxin MqsA_antitoxin SocA_Panacea ? Mac Hexapep AraC_binding HTH_18 GFO_IDH_MocA GFO_IDH_MocA_C AP_endonuc_2 XkdW ?

NZ_CP115959.1:1146861-1156105
Virgibacillus salarius

DUF4870 DUF3221 ? TetR_N YolD ? MqsA_antitoxin SocA_Panacea MqsR_toxin ? ? ? Holin_SPP1 ? ? ?

NZ_AUKH01000010.1:1226-12885
Gillisia sp. JM1

MqsA_antitoxin SocA_Panacea MqsR_toxin ? Peptidase_M64 MutS_V MutS_III ? ? ? HTH_45 TIR LRR_14 LRRs

NZ_KE384218.1:2391819-2401367
Gillisia sp. JM1

MqsA_antitoxin SocA_Panacea MqsR_toxin ? ? DUF4357 GIY-YIG ? ? TIR LRR_14 LRRs

NZ_MUGV01000012.1:113719-17126
Flavobacterium frigidimaris

PP-binding NrtR_WHD NUDIX GntR Peripla_BP_3 Aldose_epim GHMP_kinases_C GHMP_kinases_N GalKase_gal_bdg GalP_UDP_tr_C GalP_UDP_transf SSF MqsA_antitoxin SocA_Panacea FAD_binding_3 Lyase_aromatic WG WG WG WG WG WG WG WG adh_short Ketoacyl-synt_C ketoacyl-synt Lip_A_acyltrans MqsR_toxin

NZ_AUEY01000090.1:9202-14741
Desulforegula conservatrix Mb1Pa

TetR_C_13 TetR_N ? ? MqsR_toxin ? ? ORF6N DUF3606 MethyltransfD12 ? GpA_nuclease MqsA_antitoxin SocA_Panacea

NZ_AUID01000014.1:4597-22362
Proteocatella sphenisci DSM 23131

Rubrerythrin ? Pyr_redox_2 MCPsignal dCache_1 ? ? Na_H_Exchanger Lyase_aromatic Radical_SAM_N Radical_SAM DUF3362 MqsA_antitoxin SocA_Panacea Peptidase_S11 Peptidase_S26 MqsR_toxin ASCH SoFic-like_C SLT

NZ_LT906465.1:2291423-2307592
Chryseobacterium taklimakanense

? HSDR_N_2 N6_Mtase ? ? putAbiC TPRs HEPN_LA2681 MqsR_toxin NERD ? MFS_1 DUF3810 ? DUF772 DDE_Tnp_1_6 MqsA_antitoxin SocA_Panacea

NZ_CP076532.1:2556707-2565283
Bacillus sp. JNUCC-24

PBSX_XtrA Phage_integrase LinB-like_C ? Trypsin ? DUF1492 MqsA_antitoxin ? ? PAPS_reduct ? ? ? ?

NZ_BJEY01000005.1:90339-105016
Listeria newyorkensis

Alpha-amylase Glyco_hydro_38N Alpha-mann_mid Glyco_hydro_38C Glyco_hydro38C2 PTS_EIIA_2 PTS_IIB PTS_EIIC SIS HTH_6 Methyltransf_31 ? Y_phosphatase3 ? Hydrolase_3 DUF1433 HAUS5 Lipase_3 ? MqsA_antitoxin DUF1433

NZ_BBJY01000033.1:16006-11104
Vibrio diazotrophicus NBRC 103148

DUF3391 HD_5 PAS EAL DIOX_N 2OG-FeII_Oxy MqsA_antitoxin SocA_Panacea HTH_36 MarR ? ? MqsR_toxin Phage_pRha ANT ? ?

NZ_CP113865.1:451495-465954
Caldicellulosiruptor morganii

? GntR FCD Iso_dh Aconitase Aconitase Aconitase_C Peripla_BP_6 BPD_transp_1 SBP_bac_11 Nitro_FeMo-Co Radical_SAM Oxidored_nitro Oxidored_nitro HMGL-like MqsR_toxin ? MqsA_antitoxin SocA_Panacea

NZ_JYNH01000092.1:6890-18806
Desulfosporosinus sp. I2

CbiA ? ? MqsA_antitoxin SocA_Panacea MqsR_toxin Peptidase_A24 CBP_BcsQ ? Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 ? ParA-Soj-NTPase NT5C ?

NZ_JACSMY010000006.1:7831-21123
Exiguobacterium sp. s46

ParE_toxin ? TrbC DUF5388 ParA-Soj-NTPase ? Resolvase HTH_7 UvrD_C UvrD-helicase PriCT_1 MqsR_toxin Peptidase_M66 Peptidase_M43 MqsA_antitoxin SocA_Panacea HTH_3 OLD-like_TOPRIM ABC-ATPase ?

NZ_CP133671.1:7274888-7298361
Aureispira sp. CCB-E

Ribosomal_L12_N Ribosomal_L12 RNA_pol_Rpb2_2 RNA_pol_Rpb2_2 RNA_pol_Rpb2_3 RNA_pol_Rpb2_45 RNA_pol_Rpb2_6 RNA_pol_Rpb2_7 RNA_pol_Rpb1_1 RNA_pol_Rpb1_2 RNA_pol_Rpb1_2 RNA_pol_Rpb1_3 RNA_pol_Rpb1_4 RNA_pol_Rpb1_5 Synaptojanin DUF6576 Rhomboid Rhomboid Rhomboid Hydrolase ? ? HTH_3 SocA_Panacea ? Transposase_20 DEDD_Tnp_IS110 Por_Secre_tail PolyA-Pol MTPAP-like_central 2H Synaptojanin MJ1316

NZ_CABKVB010000009.1:2649103-2663541
Numidum massiliense

HTH_21 rve rve_3 HTH_Tnp_1 ? SRAP DUF3006 Lactamase_B Beta-lactamase DUF3471 DUF3471 MqsA_antitoxin SocA_Panacea MqsR_toxin TetR_N TetR_C_29 ABC_tran DUF4926 UPF0236 ABC2_membrane_3 ABC_tran

NZ_LKHP01000017.1:11312-20899
Caloramator mitchellensis

2OG-FeII_Oxy_6 DUF5651 ? ? ? NYD-SP28_assoc HTH_7 MqsA_antitoxin SocA_Panacea Terminase_2 Terminase_3 Terminase_3C SPP1_portal Phage_min_cap2 Ntox50 ? SKA1 MqsR_toxin PDDEXK_15 ?

NZ_JGYD01000011.1:81628-92135
Dehalococcoides mccartyi

PSRP-3_Ycf65 SMI1_KNR4 SMI1_KNR4 ? ? Acetylase ? MqsA_antitoxin SocA_Panacea ? AgrB Peptidase_MA_2 ? HNH ? YjzC ?

NZ_MASS01000069.1:11037-7401
Desulfosporosinus sp. BG

MqsA_antitoxin SocA_Panacea FmiP_Thoc5 T5orf172 ? HTH_Tnp_1 HTH_Tnp_1 DUF3786 FMN_red HTH_23 HTH_21 rve MqsR_toxin DUF1254

NZ_FNGB01000013.1:137-6378
Halanaerobium congolense

MazE_antitoxin ? ParE-like_toxin ? ? MqsA_antitoxin ParA-Soj-NTPase GNVR DUF7603 Wzz DUF4351

NZ_LT608335.1:224663-236656
uncultured Sporomusa sp.

BPD_transp_1 ABC_tran DUF1847 BPD_transp_1 Aminotran_1_2 Peptidase_C39_2 VTT_dom MqsR_toxin MqsA_antitoxin SocA_Panacea LysE PMSR Lactamase_B ? ? ?

NZ_LWBP01000236.1:208-9833
Niastella populi

Arm-DNA-bind_5 Phage_int_SAM_5 Phage_integrase MqsR_toxin HTH_3 SocA_Panacea ? ? DUF6161 ? HTH_17 Zn_ribbon_DnaG Toprim_2

NZ_AP043707.1:693244-708259
Paenibacillus thiaminolyticus

YkyA ? ROK Aldolase FlxA Transglutaminase MFS_1 MFS_1 sCache_3_2 HAMP HisKA_3 His-Kinase Receiver GerE ABC_tran SBP_bac_3 Abhydrolase_1 MqsA_antitoxin SocA_Panacea MqsR_toxin Phosphonate-bd

NZ_NJDE01000025.1:238214-253040
Saccharibacillus sp. O23

Glyoxalase ASCH DinB_2 Acetylase YezG-like DUF4145 HIT MqsR_toxin MqsA_antitoxin SocA_Panacea DUF6932 ? GyrI-like ABC_tran_CTD ABC_tran ABC_tran_Xtn ABC_tran TetR_C_31 TetR_N ?

NZ_FZOQ01000021.1:64828-85744
Pontibacter ummariensis

ThiF DUF6791 ? ? SocA_Panacea ? REase N6_Mtase Type_ISP_C ? Piwi Recombinase Resolvase DUF3800 ? DUF6088 DUF6527 CBM96 PGAP1 PKD_6 Por_Secre_tail ?

NZ_FOLE01000003.1:261239-288086
Flexibacter flexilis DSM 6793

MGH1-like_GH CNNM CBS CBS CorC_HlyC Calcineurin Por_Secre_tail DUF285 DUF285 DUF285 DUF285 DUF285 ? PcrA_UvrD_tudor UvrD_C UvrD-helicase MqsR_toxin MqsA_antitoxin SocA_Panacea ABC-3 CarbopepD_reg_2 Plug SusD-like_2 GDC-P GDC-P GcvP2_C Acetylase Acetylase OmpA PD40 Chorismate_bind Anth_synt_I_N

NZ_PQBS01000039.1:1401-19013
Gracilimonas amylolytica

? REase ResIII Helicase_C EcoEI_R_C ? HsdM_N N6_Mtase Methylase_S Methylase_S MqsR_toxin MqsA_antitoxin SocA_Panacea ? ? ? ? ? zf-C2H2 BACON ?

NZ_PYWA01000019.1:3485-5348
Paracoccus indicus

Rep3_C Rep3_N MqsA_antitoxin SocA_Panacea MqsR_toxin SlyX HTH_17 ? ? ?

NZ_CP084384.1:89003-106251
Lactobacillus helsingborgensis

HTH_28 HTH_28 GntR FCD Ldh_1_N Ldh_1_C MR_MLE_N MR_MLE_C GntP_permease LysR_substrate HTH_1 CitMHS CitMHS FAD_binding_2 MqsR_toxin MqsA_antitoxin SocA_Panacea 2-Hacid_dh_C MR_MLE_N MR_MLE_C Lactonase Ribul_P_3_epim HTH_21 rve rve_2

NZ_QNVW01000053.1:516-14630
Chryseobacterium sp. 5_R23647

Exonuc_VII_S tRNA_anti_2 Exonuc_VII_L MqsA_antitoxin SocA_Panacea KAP_NTPase HsdM_N N6_Mtase Virulence_RhuM Methylase_S Methylase_S HATPase_c_4 HTH_24 ? REase SWI2_SNF2 T1R_D3-like EcoR124_C MqsR_toxin SNase

NZ_WYET01000004.1:140849-156155
Flagellimonas chongwuensis

? ResIII Helicase_C ? DUF6527 DUF6791 ThiF Multi_ubiq MqsA_antitoxin SocA_Panacea MqsR_toxin HTH_3 ? HTH_26 ? ? AAA_19 AAA_15 OLD-like_TOPRIM

NZ_CALKMD010000089.1:202048-223283
uncultured Parabacteroides sp.

DUF948 His-Kinase DUF169 Acetylase MqsA_antitoxin SocA_Panacea MqsR_toxin Adenine_glyco 5-nucleotidase CHD1-like_C Helicase_C SNF2-rel_dom ? ? Arabinose_bd HTH_18 Acyl_transf_3 AAA_12 AAA_11 AAA_11 GATase_2 Glu_syn_central Glu_synthase GXGXG

NZ_CACVBY010000047.1:29718-15952
Chryseobacterium fistulae

Pribosyltran ADK MMR_HSR1 GTP1_OBG Muramidase PG_binding_1 Adenylsucc_synt Helicase_C DEAD DUF6438 ABC_tran ABC_tran MqsA_antitoxin SocA_Panacea ? ? ? ParA-Soj-NTPase ?

NZ_CALGLB010000005.1:60-11441
uncultured Prevotella sp.

? PS_pyruv_trans MatE ? ? NusG Phage_integrase Phage_int_SAM_5 Arm-DNA-bind_5 SHOCT SocA_Panacea Lipocalin_3

NZ_AP018449.1:3282559-3299967
Methylomusa anaerophila

PPDK_N PPDK_N PPDK_N PEP-utilizers PEP-utilizers_C ? Xan_ur_permease Xan_ur_permease ? ABC1 MqsA_antitoxin SocA_Panacea ? Iso_dh Flavoprotein FTCD_C ? MqsR_toxin KdpD Usp DUF4118 His-Kinase His-Kinase Kinase-PPPase

NZ_SEIZ01000002.1:235345-243474
Bacillus sp. 005/A4HT-01/001

? FAM186A-B_N ? Big_14 ? ? MqsA_antitoxin LinB-like_C DUF1492 Phage_integrase GIY-YIG ? IL10 ? ?

NZ_SPQQ01000003.1:711953-727501
Desulfosporosinus fructosivorans

ABC2_membrane_2 ABC2_membrane_2 DrrA1-3_C ABC_tran Na_H_antiporter ? 4HBT GPP34 MqsR_toxin HTH_31 SocA_Panacea SSF AcetylCoA_hydro AcetylCoA_hyd_C PAS NtrC-AAA AAA_lid_14 HTH_8 FAS1_DH_region ECH_1 Abhydrolase_1 Abhydrolase_1

NZ_AP019697.1:1691330-1706602
Dialister hominis

NTP_transferase DUF5986 Peptidase_M78 HTH_3 SecB MqsA_antitoxin MqsR_toxin DDE_Tnp_1 Glycos_transf_2 NTP_transferase Polysacc_synt Glycos_transf_2 Glyco_transf_4 Glyco_trans_1_4 Wzy_C Glycos_transf_2 Glycos_transf_2

NZ_CANMXD010000007.1:79909-97832
uncultured Planococcus sp.

DUF6290 ? UvrD-helicase UvrD_C PDDEXK_1 N6_N4_Mtase ResIII RE_endonuc Phage_integrase ? MqsA_antitoxin SocA_Panacea SMODS-polBeta ? SAVED ? ? Resolvase

NZ_JAAEEH010000021.1:18453-10327
Anaerotalea alkaliphila

UPF0236 SecB MqsA_antitoxin MqsR_toxin NTP_transferase Zn_Ribbon_1 ? SBP_bac_6 ? ABC_tran ?

NZ_JAAKEB010000005.1:60007-74302
Adlercreutzia sp. ZJ141

Pribosyltran GATase_7 Thiamine_BP HipA_C Aminotran_5 PALP TMP-TENI MqsA_antitoxin SocA_Panacea ? Pecanex_C RHH_5 PIN TetR_N ABC2_membrane_2 ABC_tran HK NifU_N

NZ_JACDUU010000014.1:12681-26101
[Anoxybacillus] calidus

DUF1232 Replic_Relax ? ? DndB ? MqsA_antitoxin SocA_Panacea MqsR_toxin ? Acetylase PHO4 PHO4 ? cobW ATP-grasp_4 LAL_C2 Phage_integrase

NZ_FYAM01000023.1:825301-839854
Clostridium sp. BSD9I1

Radical_SAM DUF434 DUF5616 ? CW_binding_2 CW_binding_2 CW_binding_2 FMN_red MqsA_antitoxin MqsR_toxin Abhydrolase_1 ABC_tran oligo_HPY ABC_tran oligo_HPY OppC_N BPD_transp_1 BPD_transp_1_N BPD_transp_1 SBP_bac_5 TerC Amidase_3

NZ_JAGKSQ010000005.1:258018-273520
Halalkalibacter suaedae

PALP DUF326 DUF326 DUF326 Coat_F ? GntR Peripla_BP_3 Aldolase_II MqsA_antitoxin SocA_Panacea ? LUD_dom LUD_dom Fer4_8 LutB_C CCG CCG HTH_DeoR DeoRC RhaA FGGY_N FGGY_C Aldolase_II adh_short

NZ_CDVNZT010000002.1:4438-19314
Salegentibacter mishustinae

VapE-like_dom DUF3874 GIY-YIG I-HmuI_NUMOD-like ? HsdM_N N6_Mtase Methylase_S Methylase_S MqsR_toxin MqsA_antitoxin SocA_Panacea HSDR_N_2 ResIII Helicase_C EcoEI_R_C ? ? Macro NAPRTase_N NAPRTase NAPRTase_C Isochorismatase ? NUMOD1

NZ_CAKKNA010000010.1:30797-44857
Bacillus sp. CECT 9360

Peripla_BP_6 BPD_transp_2 BPD_transp_2 ABC_tran BCA_ABC_TP_C ABC_tran ? MqsA_antitoxin SocA_Panacea Glyoxalase YusW MFS_1 MFS_1 Wzz GNVR ParA-Soj-NTPase CpsB_CapC Peptidase_C15 SUKH_6

NZ_CP089981.1:5724613-5737848
Fulvivirga ulvae

Phage_T4_gp19 Phage_T4_gp19 Phage_sheath_1 Phage_sheath_1C ? Pvc16_N ? MqsA_antitoxin SocA_Panacea ? Abhydrolase_1 HTH_18 GAF Ceramidase_alk Ceramidase_alk ? ? PRC Dodecin

NZ_JAKIRP010000004.1:307039-334771
Chitinophaga filiformis

IMPDH CN_hydrolase LNT_N DLH Abhydrolase_1 CHU_C Ig_7 PKD_4 PKD_4 CHU_C PKD_4 PKD_4 adh_short HTH_3 SocA_Panacea MqsR_toxin BNR_2 CarbopepD_reg_2 Plug TonB_dep_Rec_b-barrel ? SprA_N SprA_N RuvA_N HHH_5 RuvA_C ACP_syn_III_C ACP_syn_III TPRs DUF6377 His-Kinase

NZ_CP091787.1:269957-291937
Prevotella sp. E13-17

DAP_epimerase DAP_epimerase Aminotran_1_2 GSIII_N Gln-synt_C Gln-synt_C-ter ? Fic-like_C HATPase_c_4 SLFN_AlbA_2 ? MqsA_antitoxin SocA_Panacea AtzG-like ABC_tran ? DUF150_C RimP_N KH_NusA_C KH_5 S1 NusA_N GTP_EFTU_D2 IF-2 EF-G_D2 GTP_EFTU IF2_N DUF3298 PdaC Peptidase_S41 PDZ_6 CtpB_N-like

NZ_JAKZGS010000006.1:54064-71078
Belliella calami

Orn_Arg_deC_N NAD_binding_9 PALP adh_short_C2 Topoisom_I Top1B_N_bact Alpha-amylase_C Alpha-amylase Alpha-amylase CBM_48 FRG ? MqsA_antitoxin SocA_Panacea AAA_14 DUF4143 ? LytTR PRA-PH PRA-CH His_biosynth His_biosynth

NZ_JALKBZ010000002.1:17298-37268
Fuchsiella alkaliacetigena

Usp Usp CZB HAMP MCPsignal ACR_tran HlyD_D23 OEP OEP TetR_N TetR_C_24 MqsA_antitoxin MqsR_toxin Porin_4 Porin_4 ? Porin_4 ? CTP_transf_like Cons_hypoth95 RecG_wedge DEAD Helicase_C RecG_dom3_C

NZ_CAKOEW010000002.1:101519-115324
Convivina intestini

Adenylsucc_synt IMPDH HAD_2 DJ-1_PfpI ADH_N ADH_zinc_N Nitroreductase HTH_3 SocA_Panacea MqsR_toxin FAD_binding_4 Lact-deh-memb Peptidase_M24 ? TetR_N Abhydrolase_1 GATase Anth_synt_I_N Chorismate_bind

NZ_JAMXLA010000001.1:1138446-1149410
Flavobacterium sp. NRK1

HTH_3 ? DUF3127 ? ? DUF1376 DUF6633 MqsR_toxin MqsA_antitoxin SocA_Panacea HTH_Tnp_1_2 Terminase_3 ? ? ? ?

NZ_CANMVV010000068.1:235-8783
Aquimarina algiphila

Receiver LytTR Hydrolase_4 MqsA_antitoxin SocA_Panacea MqsR_toxin His_kinase ? ? Hydrolase_4 Abhydrolase_1 ?

NZ_CAJKTF010000046.1:5091-4326
Gemmiger formicilis

RepA_N DUF6076 DUF2971 SecB MqsA_antitoxin MqsR_toxin

NZ_CAKUPO010000002.1:140864-152978
uncultured Alistipes sp.

DUF3575 DUF4906 DUF4906 Mfa2 LPAM_1 ? MqsA_antitoxin SocA_Panacea ? T2SSF T2SSF ? ? ? ? ? ?

NZ_CDWLDZ010000059.1:1273-8802
Megamonas hypermegale

MqsR_toxin RBFA YlxR NusA_N S1 KH_5 KH_NusA_C IF2_N IF2_N GTP_EFTU EF-G_D2 IF-2 GTP_EFTU_D2 TruB_N TruB_C_2 TruB-C_2 Ribosomal_L7Ae DHH DHHA1 MqsA_antitoxin

NZ_JAUFQQ010000005.1:2191724-2209782
Flavobacterium branchiarum

? TatD_DNase Acyltransferase Asparaginase Asparaginase_C MFS_3 HTH_1 LysR_substrate MqsR_toxin MqsA_antitoxin SocA_Panacea HTH_3 Acetylase CN_hydrolase HTH_18 DUF6597 Isochorismatase Acetylase SoFic-like_C Fic Fic_N

NZ_JAGDWY010000015.1:20388-31390
Desulfurella sp.

GIY-YIG UVR UvrC_RNaseH UvrC_RNaseH_dom HHH_5 Pyr_excise ? AAA AAA_lid_3 RNase_Y_N ? PAS GGDEF MqsR_toxin MqsA_antitoxin SocA_Panacea ? ? UPF0102 PEPCK_N

NZ_AP026678.1:2584292-2608312
Ignavibacterium sp.

Aminotran_5 DUF1015 Esterase Acid_phosphat_B Eco57I N6_Mtase ? MqsA_antitoxin SocA_Panacea ? Val_tRNA-synt_C Anticodon_1 tRNA-synt_1 DinB Malt_amylase_C Alpha-amylase AMPK1_CBM ? Por_Secre_tail Sortilin-Vps10 Sortilin-Vps10 Por_Secre_tail FG-GAP_2 FG-GAP_2 FG-GAP_2 Por_Secre_tail SBBP SBBP SBBP SBBP

NZ_CASDUT010000006.1:14583-32226
uncultured Bacteroides sp.

AMP-binding_C AMP-binding PP-binding GNVR SLBB SLBB SLBB_2 SLBB SLBB_2 Poly_export Bac_transf CoA_binding_3 ? BT4734-like_N DUF3987 ? MqsA_antitoxin SocA_Panacea DUF4248 HU-HIG Amidase_2 CoA_binding_3 ? ?

NZ_CP133721.1:1645072-1661189
Flavobacterium nakdongense

Mur_ligase_C Mur_ligase_M ? zf-CGNR ? HsdM_N N6_Mtase MqsA_antitoxin SocA_Panacea MqsR_toxin Acetylase Methylase_S Methylase_S DUF5131 GMT-wHTH REase ResIII Helicase_C EcoEI_R_C DUF5655 ? Resolvase Recombinase Zn_ribbon_recom

NZ_JAMQKA010000059.1:11234-12299
Aliifodinibius sp. S!AR15-10

Phage_AlpA ? MqsA_antitoxin SocA_Panacea ? N6_Mtase Type_ISP_C Zn_ribbon_16 DUF6371 D5_N DUF5906 Pox_D5 TPR_MLP1_2 DUF4566 Arm-DNA-bind_3 Phage_int_M Phage_integrase Sigma70_r4_2 Sigma70_r2

NZ_DAMBUX010000005.1:178962-201218
Sedimentibacter saalensis

SLH SLH YDG YDG YDG YDG YDG YDG YDG YDG YDG YDG YDG YDG Wzy_C CoA_binding_3 Polysacc_synt_2 FMN_bind FAD_binding_2 FAD_binding_2 SecB MqsA_antitoxin MqsR_toxin Peptidase_M20 Peptidase_M20 MFS_1 DcuC TetR_N GDP_Man_Dehyd Glycos_transf_2 TPRs TPRs

NZ_JAYGIM010000019.1:41213-29405
Arcicella lustrica

DUF7619 RHS_repeat RHS_repeat RHS_repeat RHS_repeat RHS_repeat RHS_repeat RHS_repeat DUF6531 CARDB CARDB CARDB CARDB CARDB CARDB CARDB CARDB fn3 Beta_helix Patatin REase SWI2_SNF2 T1R_D3-like N6_Mtase PBECR3 Methylase_S Methylase_S MqsR_toxin HTH_3 SocA_Panacea ? PycTM Guanylate_cyc ? Fic_N Fic HxlR

NZ_BAABCR010000003.1:26613-44375
Flavobacterium cheonhonense

? ? DUF2971 TIR DUF6943 DUF4238 MqsR_toxin MqsA_antitoxin SocA_Panacea AAA_16 ? HTH_3 ? SMC_N DUF5675 ?